

CHAPTER 1

INTRODUCTION

1.1 BACKGROUND

ARTIFICIAL INTELLIGENCE SHOULD ALSO PROVIDE SOLUTIONS IN OVERCOMING PROBLEMS

INDONESIA is a strategic country in the form of an archipelago with a great number of population with a diversity of cultures and local wisdom, as well as has economic growth that continues to increase from year to year. Indonesia has a million opportunities in utilisation of Artificial Intelligence (AI) because the technology can potentially provide growth in productivity for businesses, investment efficiency in the employment of human resources, and innovation in various sectors, such as finance, health, education, agriculture, defence and security, transportation, and maritime affairs.

Artificial intelligence can also provide cost-effective solutions to address infrastructure problems, deliver effective social services, plan the quality of educational resources, support the governments in formulating the appropriate policies, build convenient digital markets, assist governments in providing the best services in public sectors, and so forth.

However, Indonesia as a country that currently has the largest power in the digital economy market in Southeast Asia will face various challenges in implementing the technology. These challenges can be grouped into four important matters, namely readiness of skilled talents who develop and use AI, readiness of regulations governing the ethical application and AI utilisation in a responsible way, readiness of computational infrastructure and supporting data for AI modelling, readiness of the corporations and public sectors in adopting AI innovations.

To face the opportunities and challenges mentioned above, the government of Indonesia needs to arrange an Indonesian artificial intelligence strategy by taking into account the issues that exist in the artificial intelligence strategies of other countries, both globally and regionally.

The government of Indonesia is also considering aligning the vision and mission of Indonesia's artificial intelligence with national interests, as well as reviewing Indonesia's SWOT with the aim to get the accurate focus areas and sectors' priority.

NATIONAL STRATEGY ON ARTIFICIAL INTELLIGENCE TECHNOLOGY

- National Strategy of Artificial Intelligence of Countries around the World

Various countries in the world already have a national strategy on artificial intelligence as a reference for stakeholders in respective countries in implementing artificial intelligence technology and understanding the direction of the government's national policies in the field of artificial intelligence. Consequently, the government of Indonesia must determine its own national strategy on artificial intelligence so that Indonesia is capable of competing and must not be left behind in taking opportunity of the development and maximum utilisation of artificial intelligence technology for the people of Indonesia. The national strategy on artificial intelligence is required so that the development and utilisation of artificial intelligence technology can be in harmony with national interests and have ethical responsibilities whose values are based on Pancasila. In addition, the government must also prioritise the work of the nation's youth to be adopted by the industrial and public sectors.

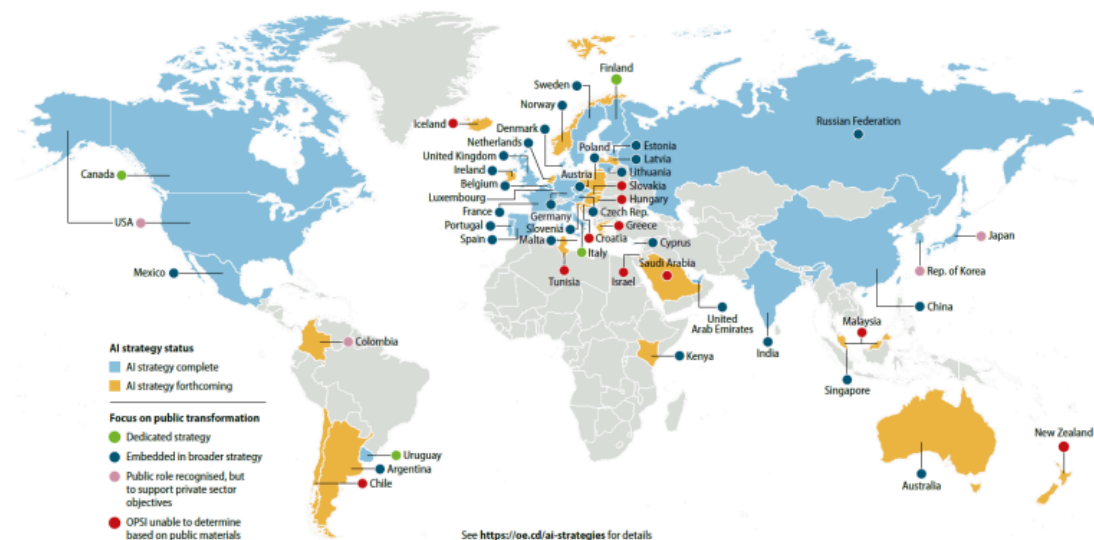


Figure 1.1 Implementation landscape of National Strategy on Artificial Intelligence in various countries [1]

The landscape of the deployment of countries that already have a national strategy on artificial intelligence is shown in Figure 1.1, explaining that global countries have developed a national strategy on artificial intelligence as a response to opportunities and challenges for their national interests. The German Federal

**THE GOVERNMENT OF INDONESIA
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ARTIFICIAL INTELLIGENCE
THROUGH A HOLISTIC
APPROACH.**

Government launched its vision with the slogan "AI Made in Germany" [2], the Chinese Government with "Deciphering China's AI Dream" [3] and the South Korean Government with "Toward AI World Leader beyond IT" [4]. The results of the study from Konrad Adenauer Stiftung [5] on the national strategies of six leading countries (United States, China, The Great Britain, France, Finland, and South Korea) show that there are several weaknesses in the national strategies of these countries, including: (1) the definition of artificial intelligence is still vague and inconsistent; (2) setting goals that are unclear, making it difficult to measure success; (3) emphasise more on the strategic aspects of data availability and specialist training than on the expansion of domestic computing capacity; and (4) setting aside solutions to create an ecosystem.

The Indonesian government is attempting to arrange a national strategy on artificial intelligence through a holistic approach, having a clear definition of artificial intelligence, defining measurable goals, and considering solutions in the form of ecosystems that can orchestrate the entirety of Indonesia's strengths and potential. The strategy created must be feasible and effective for the progress of the nation. In addition, Indonesia will establish a national slogan for artificial intelligence that can display the confidence of artificial intelligence technology for the success of Vision Indonesia 2045.

• National Strategy on Artificial Intelligence of Regional Countries

Regionally, Singapore is the fastest and most advanced country in implementing artificial intelligence. Singapore released the National AI Strategy (NAIS) which arranges artificial intelligence planning, as a practicable lead to the future intelligent nation. Singapore established the National AI Office (NAIO) as an institution that coordinates all the creation of various artificial intelligence talents and prevents factions from working in silos.

Initially, Singapore defined four main artificial intelligence initiatives: (1) Fundamental AI Research which funds scientific research that will contribute to other pillars of Singapore's Artificial Intelligence; (2) Grand Challenges which support the work of multi-disciplinary groups that supply innovative solutions to the key challenges faced by Singapore and the world. Currently the program focuses on health, urban solutions and finance; (3) 100 Experiments which funds scalable artificial intelligence solutions to

problems identified by industries; and (4) AI Apprenticeship is a 9-month structured program to cultivate a new pool of AI talent in Singapore.

Furthermore, in June 2018, the government of Singapore announced three new initiatives on the governance and ethics on artificial intelligence. In this case, Singapore established a new Advisory Council on Ethics Utilisation of AI and Data that will support the Government in developing standards and governance frameworks for artificial intelligence ethics. The huge potential of Singapore's artificial intelligence can be an opportunity for cooperation with Indonesia in various potential sectors, such as: industrial research and innovation.

- **The G20 on Artificial Intelligence Principles and Recommendations**

Obedient with the provisions of the G20 AI Principles [7] agreed at the G20 Summit meeting on 28-29 June 2019 in Osaka-Japan, the government of Indonesia will consider that the development and utilisation of artificial intelligence must adhere to the principles into an ethical artificial intelligence that is responsible and in accordance with the values of Pancasila. The principles of artificial intelligence established by the G20 are (1) Inclusive growth, sustainable development and prosperity; (2) Human-centred values and justice; (3) Transparency and clarity; (4) Robustness, safety and security; (5) Accountability.

Additionally, the government of Indonesia also followed up the recommendations directed by the G20 as the basis for government policies on artificial intelligence. These recommendations are summarised in five goals: (1) Investment in artificial intelligence research and development; (2) Fostering a digital ecosystem for artificial intelligence; (3) Shaping an enabling policy environment for artificial intelligence; (4) Building human capacity and preparing for labour market transformation; (5) International cooperation for creating trustworthy AI.

**THE DEVELOPMENT AND
UTILISATION OF ARTIFICIAL
INTELLIGENCE TECHNOLOGY
SHOULD BE PRIORITISED ON THE
FOUR PILLARS OF THE VISION OF
GOLDEN INDONESIA 2045.**

**NATIONAL ISSUES FOR THE NATIONAL STRATEGY OF
ARTIFICIAL INTELLIGENCE**

- Vision of Indonesia 2045

Vision of Indonesia 2045 [8] is a significant issue that the government must take into account in establishing national strategies for artificial intelligence. The entire artificial intelligence mission is focused on realising the Vision of Golden Indonesia. There are four main pillars of the Vision of Golden Indonesia 2045, namely: (1) Human Development and Mastery of Science and Technology; (2) Sustainable Economic Development; (3) Equitable Development; and (4) Strengthening National Resilience and Governance. Thus the development and utilisation of artificial intelligence technology must prioritise the said four pillars. Specifically, food security and governance are priority areas that must be highlighted to receive artificial intelligence support.

- National Priority Programs from RPJMN IV 2020-2024

The government of Indonesia needs to produce initiatives for the application of artificial intelligence that are in compliance with priority programs that have been planned in the 2020-2024 Rencana Pembangunan Jangka Menengah Nasional (National Medium-Term Development Plan) or RPJMN, especially for medium-term initiative programs. Thus the artificial intelligence initiative program can be a part that simultaneously receives funding from the government spending budget. The seven development agendas of RPJMN IV 2020-2024 [9] are: (1) Strengthening Economic Resilience for Quality Growth; (2) Developing Areas to Reduce Inequality; (3) Improving the Quality and Competitiveness on Human Resources; (4) Mental Revolution and Cultural Development; (5) Strengthening Infrastructure to Support Economic Development and Basic Services; (6) Building the Environment, Increasing Disaster Resilience and Climate Change; and (7) Strengthening Stability in Politics, Law, Defense, Security, and Public Service Transformation.

- Making Indonesia 4.0

The government of Indonesia's strategy in implementing artificial intelligence will align with the "Making Indonesia 4.0" initiative [10] which is the roadmap for the Ministry of Industry in the Industrial Revolution 4.0. The roadmap displays a clear direction and strategy for the movement of

Indonesian industry in the future, including in the five industrial sectors' focus and 10 national priorities in an effort to strengthen Indonesia's industrial structure. The five industrial sectors' focus are: (1) Food and beverages; (2) textiles and clothing; (3) automotive; (4) chemistry; and (5) electronics. These sectors were chosen to be the spotlight because they have the economic and feasibility impact on GDP, trade, potential impact on other industries, the amount of investment, and the speed of market penetration.

- National Research Master Plan 2017-2045

The government of Indonesia must strive for artificial intelligence program initiatives in the area of industrial research and innovation to synergize with national research priorities that have been arranged in the Rencana Induk Riset Nasional (National Research Master Plan) 2017-2045 or RIRN [11]. RIRN is a planning document that provides priority directions for science and technology development for a period of 28 years (2017-2045). RIRN determines the missions, namely: a) creating an innovative Indonesian society based on science and technology, and b) creating a global competitive advantage for the nation (Presidential Regulation No. 38 of 2018). As a further elaboration, it is necessary to create more planning in technical aspects in the form of National Research Priorities for a period of 5 years. As mandated in the 2005-2025 RPJPN (National Long-Term Development Plan), the execution of research is focused on seven areas of the National Research Excellence Program (PUNAS), namely: (1) Food security; (2) energy, new and renewable energy; (3) health and medicine; (4) transportation; (5) information and communication technology (ICT); (6) defence and security technology; and (7) advanced materials.

- National Industrial Development Master Plan 2015-2035

The government of Indonesia requires the development of artificial intelligence in industrial innovation to support the 4th industrial revolution, in accordance with Law Enactment Number 3 of 2014 [11] concerning Industry which places industry as one of the pillars of the economy and gives the government a huge role to encourage the advancement of planned national industry. The role is required in directing the national economy to grow rapidly and catch up with other countries that have initially

advanced. In order to strengthen and clarify the government's role in national industrial development, therefore a systematic, comprehensive and futuristic national industrial development plan has been compiled in the form of the 2015-2035 National Industrial Development Master Plan (RIPIN).

RIPIN 2015-2035 is a form of the government's solemnity in achieving the objectives of industrial arrangement, namely:

1. Accomplish the national industry as a pillar and driving force for national economy;
2. Accomplish the depth and strength of the industrial structure;
3. Accomplish an independent, competitive and advanced industry, as well as a Green Industry;
4. Accomplish the business certainty, fair competition, and preventing centralisation;
5. or industrial domination by a group or individual that harms society;
6. Open the business opportunities and expansion of employment opportunities;
7. Accomplish the equal distribution of industrial development to all regions of Indonesia, in order to
8. strengthening and enhancing the national resilience; and
9. Increasing the prosperity and welfare of society in a fair manner

SATU DATA INDONESIA (ONE DATA INDONESIA) BECOMES AN IMPORTANT INSTRUMENT TO BE SUCCEED BY THE GOVERNMENT OF INDONESIA IN SUPPORTING THE ACCELERATED GROWTH OF INDONESIA'S ARTIFICIAL INTELLIGENCE INNOVATION.

- **National Capital Relocation Plan**

The government is currently still reviewing and preparing every possibility of moving the National Capital of Indonesia. The government stipulates that the New Capital City (IKB) has four criteria: (1) IKB is a symbol of national identity; (2) IKB that is green, smart, beautiful, and sustainable; (3) IKB that is modern and international standard; and (4) IKB with the efficient and effective governance. To achieve these four criterias, the government of Indonesia can utilise and adopt as much as possible of artificial intelligence technology for all sectors, both public and private, especially to make the Capital as a smart and modern city.

- **One Data Indonesia (Perpres No. 39 2019)**

The development of artificial intelligence is inseparable from the learning results from an abundance of data. The more sufficient and quality the data is, the result artificial intelligence will have better performance. In this case, the President of the Republic of Indonesia, has issued an One Data Indonesia Regulation contained in Presidential Decree No. 39 2019 [12]. The Presidential Regulation controls the implementation of One Data Indonesia to be able to be shared and accessed responsibly. There are a number of regulated principles, namely: meet the standard of data, have the metadata, meet the data interoperability rules, and use reference codes and/or master data. Thus the regulation of One Data Indonesia is becoming a significant instrument for the government of Indonesia to succeed in supporting the accelerated growth of Indonesia's artificial intelligence innovation.

- **Electronic-Based Government System (Perpres No.95 2018)**

Electronic-Based Government System (SPBE) is a government enforcement that utilises information and communication technology to provide services to SPBE Users. It is stated in Presidential Regulation No. 95 of 2018 about Electronic-Based Government Systems [13]. SPBE is aimed at achieving a clean, effective, transparent and accountable governance as well as qualified and reliable public services. National-level governance and management of electronic-based government systems are also required to improve the integration and efficiency of electronic-based government systems. The vision of SPBE

is "the achievement of an integrated and comprehensive electronic-based government system to attain high-performance bureaucracy and public services." In order to accomplish the vision of SPBE, the artificial intelligence technology can be studied for its utilisation to such an extent in helping the government generate accurate decisions and policies.

- **COVID-19 Pandemic**

The pandemic caused by COVID-19 is the first global health crisis in this century. The government of Indonesia boosts and facilitates artificial intelligence in providing solutions to problems that arise, in terms of detecting, analysing and managing every possible scenario caused by the pandemic. A series of innovations on artificial intelligence applications that can be developed to overcome the COVID-19 pandemic are early detection and diagnosis of infection, patient care monitoring, contact tracing, projection of cases and deaths, medicine and vaccine development, reduction of workloads for the health workers, and disease prevention. disease. Several rapid projects that mobilise the advantages of artificial intelligence can be supported and implemented by the government of Indonesia.

1. 2 SWOT ASSESSMENT AND NATIONAL STRATEGY ON ARTIFICIAL INTELLIGENCE

THE INDONESIA'S NATIONAL SWOT ASSESSMENT ALSO CONSIDER ACTUAL ISSUES

Table 1.1 Strength, Weakness, Opportunity, and Threat (SWOT)

Indonesia's National SWOT on artificial intelligence is assessed based on the parameters formulated in the framework of national strategy development on artificial intelligence from the World Economic Forum. The parameters used for Strengths and Weaknesses are: (1) Manpower; (2) Digitalization/Infrastructure; (3) Industrial Collaboration - Universities or Research Institutions; (4) Training Capacity; and (5) Regulation. Meanwhile, the parameters for Opportunities and Threats are: (1) Innovation Ecosystem; (2) Industry Adoption; (3) Adoption of Public Sector; and (4) International Collaboration.

Indonesia's National SWOT Assessment also considers the actual issues raised in the previous section, which are part of Indonesia's opportunity and challenges in implementing artificial intelligence.

STRENGTH	WEAKNESS
MANPOWER	
- Indonesia's Demographic Bonus, which can be used as a practicable and exceptional talent in various sectors for the development and utilisation of artificial intelligence. - The new generation of Indonesia are increasingly interested in studying Information Technology, Computer Science, and Science which is a requirement for mastering artificial intelligence.	- The society of Indonesia has not been properly educated about the knowledge and learning of artificial intelligence. - Lack of ability for graduates of mathematics, science, engineering, and computer science to be ready to dive into projects on Artificial Intelligence. - There are still a small number of experts, lecturers and professors in the field of artificial intelligence in Indonesia.
DIGITALISATION/INFRASTRUCTURE	
Bahasa Indonesia as the Language of Unity and also the Bahasa Daerah (local language) which are valuable national assets for the Bahasa-based Artificial Intelligence modelling dataset.	There is still a minimum number of unique data sets belonging to Indonesia that are open and can be shared for research and experimentation, such as: the Indonesian language corpus, both text and speech.

STRENGTH	WEAKNESS
DIGITALISATION/INFRASTRUCTURE	
• The number of potential unique datasets originated from cultural diversity, local wisdom, and various sectors of the life of the Indonesian people, in particular: agriculture, government, disaster, maritime, and others. • Adequate telecommunications and internet networks for cloud computing services • Development of national infrastructure and connectivity which is improving throughout the archipelago. • Indonesia is the largest digital economy market in Southeast Asia with a potential for growth of four times by 2025.	• The regulation implementation of One Data Indonesia still experiences silos or ego-sectoral attitudes from each data owner. • Network security systems and data information are still easily hacked. • Inadequate digital infrastructure in universities and research institutions to conduct research and experimentation on the development of artificial intelligence. • Digital datasets are scattered and the data in quantity and quality still needs to be improved for experimentation purposes.
COLLABORATION OF INDUSTRY, UNIVERSITIES, AND RESEARCH INSTITUTIONS	
• The government provides incentive funds for research programs resulted from industrial collaboration with universities or government research institutions, such as: RISPRO LPDP (RISPRO Educational Fund Management Institution) and Riset Pro of the Ministry of Research and Technology.	• The link and match harmonisation between the world of education and local industries is still not optimal, such as: internship opportunities and student final thesis directly in the industrial field, and the availability of research topics for final thesis and research that can be undertaken by universities. • Investment funding in research and education are still low, especially those from industry. • There is still a small amount of research and development on artificial intelligence innovations carried out by collaboration between industry and universities or research institutions. • The quality of products resulted from research innovations from universities or research institutions have not been well adopted by the industry.
TRAINING CAPACITY	
• Indonesia has universities that are escalated evenly throughout the archipelago and some are universities with international standards.	• Most tertiary institutions do not yet have adequate requirements to support artificial intelligence learning, such as: competent teaching staff in the field of artificial intelligence, laboratories, datasets, learning syllabus, and others. • The school education system has not yet introduced the field of artificial intelligence in the education curriculum.

STRENGTH	WEAKNESS
TRAINING CAPACITY	
	• There are no incentives for specific educational/training capacity development programs for artificial intelligence start-ups. • Not enough incentives for artificial intelligence capacity development in some institutions that function in priority areas, such as: agriculture, research institutions, government agencies, and others.
REGULATION	
• Availability of a legal canopy for the ethical utilisation and development of responsible artificial intelligence, originated from Pancasila and the 1945 Constitution. • Strong government motivation to the use of domestic innovation production through TKDN (Domestic Component Level) regulations and proposition to apply the Love of Domestic Products to the society.	• There are no regulatory instruments governing ethics and policies for the development and utilisation of responsible Artificial Intelligence in Indonesia. • There is no supervisory institution that monitors and controls the development and utilisation of artificial intelligence in society. • There are not enough regulations governing the issue of data and shared infrastructure that will be used publicly and openly by the public for research and innovation development purposes. • There is no national standard for Artificial Intelligence of Indonesia.

OPPORTUNITIES	THREATS
INNOVATION ECOSYSTEM • Implementation of a conducive innovation ecosystem to combine Investors, Government, Industry, Universities, Research Institutions, and non-governmental organisations. • The government has started implementing the One Data Indonesia Program through Presidential Decree No. 39 in 2019 for sharing purposes for the advancement of research and development of technological innovations. • VISION OF GOLDEN INDONESIA 2045: (1) Human Development and Mastery of Science and Technology; (2) Sustainable Economic Development; (3) Equitable Construction Development; and (4) Strengthening National Resilience and Governance. • Indonesia has planned national priority areas in the RPJMN, the Making Indonesia 4.0 Roadmap, and the National Research Master Plan (RIRN). INDUSTRY AND PUBLIC SECTOR ADOPTION • The Discourse on the Movement of a New Capital City which has the criteria of being smart, beautiful and sustainable, as well as modern international standards with efficient and effective governance. • The high demand for AI experts to support Industrial Revolution 4.0. • Indonesia is a country with a conducive startup ecosystem. • Electronic-Based Government System Program that requires Artificial Intelligence to support the government in making accurate policies. INTERNATIONAL COOPERATION • Research collaboration with institutions of global and regional countries to build the capacity of Indonesian talents abroad.	INDUSTRY AND PUBLIC SECTOR ADOPTION • Indonesia's digital market which is dominated by imported technology • Large-capitalisation of multinational companies dominate Indonesia's vital projects and many startups fall into their hands. • Decreased public trust in products made by the nation's people. • Misuse of technology that harms society. • Misuse of data's privacy that goes unnoticed. • Talented human resources who prefer to work abroad or work in foreign companies rather than in their home country because they get paid more. • Inconsistent legislation in its implementation. • Disruption of artificial intelligence that causes workforce shifting failures.

INDONESIA DEFINES THE FIVE PRIORITY AREAS TO SUCCEED THE MISSIONS OF INDONESIA'S NATIONAL STRATEGY ON ARTIFICIAL INTELLIGENCE

The results of the SWOT assessment led the government of Indonesia to arrange a national strategy on artificial intelligence into **Four Focus Areas**, namely:

- 1. Ethics and Policy**
- 2. Talent Development**
- 3. Infrastructure and Data**
- 4. Industrial Research and Innovation**

In addition to the focus areas, the government of Indonesia defines **Five Priority Areas** to succeed the missions of Indonesia's national strategy on artificial intelligence.

- 1. Health services**
- 2. Reformation of Bureaucratic**
- 3. Education and Research**
- 4. Food Security**
- 5. Mobility and Smart Cities**

INDONESIA'S NATIONAL STRATEGY ON ARTIFICIAL INTELLIGENCE

To achieve Vision of Indonesia 2045, the government of Indonesia is examining effective strategies for artificial intelligence that can lead Indonesia to the four conditions it aspires to:

1. **Indonesia is sovereign** because there is Indonesia's Data Sovereignty for the benefit of Indonesia and is not controlled by foreign parties.
2. **Indonesia is progressive** because of the superior innovations of artificial intelligence produced by Indonesia's centres of excellence.
3. **Indonesia is fair** because Indonesia applies ethical and responsible artificial intelligence both in its development and in its utilisation.
4. **Indonesia is prosperous** because Indonesia demands rapid economic growth with industry and the public sector assisted by artificial intelligence.

Based on the National SWOT assessment, these strategies are mapped into the artificial intelligence's four focus areas defined in the previous section.

1. Strategy for Ethics and Policy

1) Implement the ethics of data sharing according to regulations based on Pancasila and the 1945 Constitution.	11) Create a national data sovereignty policy that is strict to be used by multinational companies.
2) Construct institutional policies that orchestrate the ecosystem of national artificial intelligence innovation.	12) Strengthen laws to prosecute misapplication of technology.
3) Establish a Data Ethics Council to regulate data sharing and monitor its utilisation.	13) Strengthen laws to prosecute misuse of data's privacy.
4) Generate policies for data collection and sharing, including from One Data Indonesia for artificial intelligence research and experimentation.	14) Produce policies that prioritise products made by the nation's people in government procurement to reduce dependence on imported products as well as to increase the TKDN (Domestic Component Level).
5) Arrange an Ethical Policy on the Utilisation and Development of artificial intelligence, especially in priority areas of artificial intelligence technology.	15) Offer innovation development incentives for sustainable improvement on domestic products.
6) Arrange government policies and incentives to accelerate startup company/industry growth.	16) Establish a Data Ethics and National Artificial Intelligence Council that supervise the responsible use of Artificial Intelligence in society.
7) Create guidelines and policies for the utilisation and application of artificial intelligence for electronic-based government systems.	17) Awarding competitive human resources in the artificial intelligence field with greater salary levels in the country.
8) Produce technical guidelines on the implementation of international collaboration in the context of education, research and joint-development of artificial intelligence.	18) Political will from the executive and legislature in executing and supervising the regulations that have been decided.
9) Prepare the National Standards to support the products and systems produced by the ecosystem orchestra of national artificial intelligence innovation.	19) Prepare policies to face the shifts in employment by the presence of artificial intelligence
10) Create policies to increase public consumption's power and interest in using domestic innovative products.	20) Produce policies that prioritise products made by the nation's people in government procurement to reduce dependence on imported products as well as to increase the TKDN (Domestic Component Level).

2. Strategy for Talents Development

- 1) Maximise the potential of Indonesia's Demographic Bonus for the development of Indonesian talents on artificial intelligence who are practicable, as well as have competitiveness and good persona through formal and non-formal education.
- 2) Utilise the advantage of the number of adolescents who are interested in the field of artificial intelligence to develop superior and practicable Indonesian talents on artificial intelligence through an effective education system, in the context to succeed the Indonesian Industrial Roadmap (Making Indonesia 4.0).
- 3) Recruit artificial intelligence talents as the skilled manpower who will develop and operate artificial intelligence in national sectors' priority.
- 4) Develop a technology education program on artificial intelligence in all formal and non-formal educational institutions to produce numerous competitive Indonesian talents on artificial intelligence, starting from the elementary and secondary school levels.
- 5) Construct an artificial intelligence education system that produces artificial intelligence talents as workers, researchers and entrepreneurs in the field of artificial intelligence.

3. Strategy for Infrastructure and Data

- | | |
|--|---|
| 1) Build a cloud computing network infrastructure and provision of Indonesia's unique public datasets (such as: the corpus in Indonesian language) to support the productivity of the national artificial intelligence's innovation ecosystem. | 7) Utilise Indonesia's huge population of Internet users as Southeast Asia's largest digital economy market for Indonesian startups. |
| 2) Building cloud computing network infrastructure and provision of public datasets originated from the One Data Indonesia data centre for artificial intelligence research and experimentation. | 8) Improve the One Data Indonesia implementation system that tackles the silos and ego-sectoral attitudes of data owners. |
| 3) Maximise the use of cloud computing technology for equitable development and utilisation of artificial intelligence in all national sectors' priority development. | 9) Construct a data sharing system for artificial intelligence research and experimentation originated from the One Data Indonesia centre, which meets the standards in network security and data information, and is professionally managed. |
| 4) Leverage all of Indonesia's current advantages (infrastructure, data, funding and ecosystem) to achieve targets in Indonesia's national priority areas. | 10) Improve digital government datasets for learning material and analysis of strategic planning for the application of Artificial Intelligence for Electronic-Based Government Systems and the New Capital City of the State. |

3. Strategy for Infrastructure and Data

5) Build a cloud computing network infrastructure and provision of unique public datasets for the development of smart and modern artificial intelligence innovations for the New Capital City.	11) Improve the datasets from the industrial sector for the development of artificial intelligence innovations in the priority areas of Making Indonesia 4.0.
6) Utilise the cloud computing infrastructure to encourage contribution on artificial intelligence in priority areas of Making Indonesia 4.0.	12) Prepare the infrastructure to face the shifts in employment by the presence of artificial intelligence.

4. Strategy for Industrial Research and Innovation

1) Leverage all of Indonesia's current strategic advantages (infrastructure, data, funding and ecosystem) to achieve targets in Indonesia's national priority areas.	5) Promote the utilisation of AI products produced from the innovation of the nation's people.
2) Develop technical guidelines for the manifestation of a national AI innovation ecosystem (triple-, quadruple, penta-helix) that make link-and-match effectiveness for all stakeholders (government, industry, universities/research institutions, and society).	6) Provide incentives and more investment on innovation research in priority areas of artificial intelligence.
3) Increase the investment from both industry and the government to accelerate innovation growth that can be absorbed by industry and the public sector.	7) Utilise all of Indonesia's current strategic advantages (infrastructure, data, funding and ecosystem) to achieve targets in Indonesia's national artificial intelligence priority areas.
4) Increase the incentives for research and development of new technological innovations originated from collaboration between industry and universities/research institutions.	8) Advance the quality and quantity of innovative products that can be absorbed by the Indonesian digital market through the incubation process and product dissemination to the public.

1. 3. FRAMEWORK OF NATIONAL STRATEGY ON ARTIFICIAL INTELLIGENCE

THIS FRAMEWORK ASSISTS TO DESCRIBE THE EXISTENCE OF PRIORITY AREAS AND THE RELATIONSHIP BETWEEN NATIONAL INITIATIVE PROGRAMS

THE GOVERNMENT of Indonesia developed an Indonesian Framework of National Strategy on Artificial Intelligence with the aim to provide guidance on the interrelationships of the four focus areas and assigned missions. In addition, the framework helps to describe the existence of priority areas, and the linkages of national initiative programs from the four focus areas with priority areas and the selection of Quick-Wins activities for the use of artificial intelligence. The Indonesian Framework of National Strategy on Artificial Intelligence is shown in Figures 1-2.

Figure 1-2. The Indonesian Framework of National Strategy on Artificial Intelligence

